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RESULTS

Results

01 · ONE-SAMPLE T-TEST

does a mean differ from a fixed value?

Table 1. Descriptives

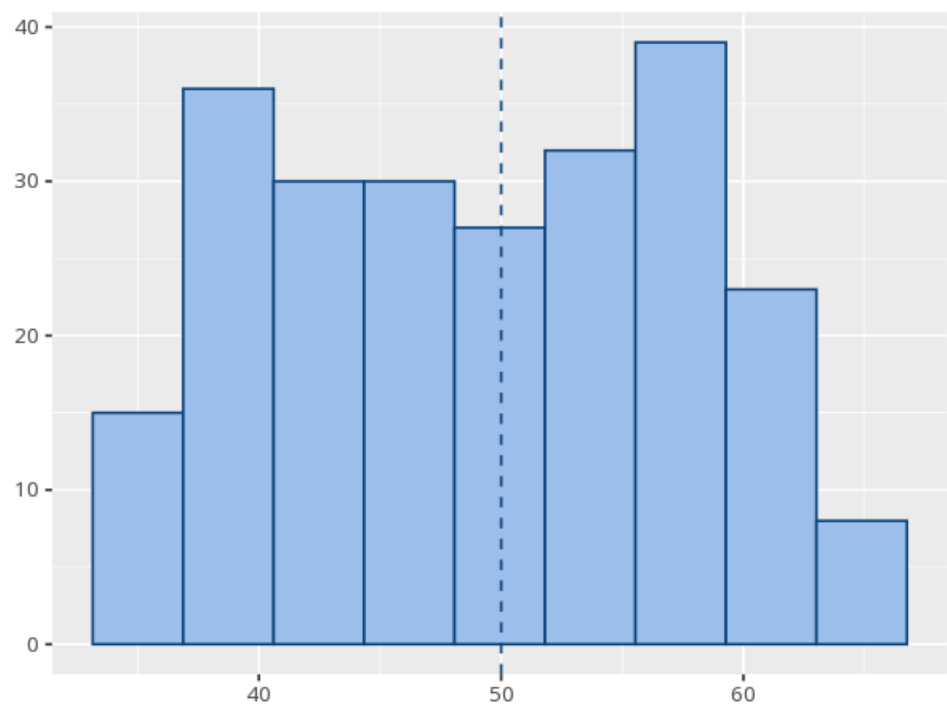
Variable	N	M	SD	SE
pretest_score	240	49.25	8.46	0.55

Table 2. One-sample t-test

Test value	t	df	p	M _{diff}	95% CI	Cohen's d [95% CI]
50	-1.38	239	.170	-0.75	[-1.83, 0.32]	-0.09 [-0.22, 0.04]

assumption check: normality (Shapiro-Wilk) reported under the descriptives. (Shapiro-Wilk W=0.95, p=<.001) — normality looks doubtful; interpret the effect size and CI with extra caution

Figure. Value vs. test value



Understanding the numbers

CI. The range of plausible values for the size of the difference between your sample mean and the test value. Here, the difference's CI is $[-1.83, 0.32]$.

Cohen's d. The size of the difference in standard-deviation units (~ 0.2 small, 0.5 medium, 0.8 large). Here, $d = -0.09$ $[-0.22, 0.04]$.

df. The degrees of freedom for the t-test, based on your sample size. Here, $df = 239$.

M diff. How far your sample mean sits from the test value. Here, $M \text{ diff} = -0.75$.

M. The average value of your sample. Here, $M = 49.25$.

μ_0 . The fixed value your sample mean is being compared against. Here, $\mu_0 = 50$.

N. The number of observations used in the test. Here, $N = 240$.

p. The probability of a mean this different from the test value (or more so) if the true mean equaled it. Here, $p = .170$ - at or above the conventional $.05$ threshold, no significant difference detected.

SD. How spread out your sample values are around the mean. Here, $SD = 8.46$.

SE. The precision of the sample mean as an estimate of the true population mean. Here, $SE = 0.55$.

t. How many standard errors your sample mean is from the test value. Here, $t(239) = -1.38$.

Variable. Which column was analyzed. Here, the variable analyzed is `pretest_score`.

How to read this test

Compares your sample mean to a fixed value. A p below α means the mean differs from that value; the mean difference and 95% CI show by how much (its sign tells you whether the mean is above or below the test value), and Cohen's d gives the effect size. A p at or above α does not prove the mean equals the test value — only that you lack evidence it differs. Your significance threshold (α) is 0.05 .

APA template: A one-sample t-test gave $M=49.2$ vs. 50 , $t(239)=-1.38$, $p = .170$, $d=-0.09$ $[-0.22, 0.04]$.

Statistical basis: One-sample t-test (Student (1908). "The probable error of a mean." *Biometrika*, 6(1), 1-25.); Cohen's d effect-size benchmarks (.2/.5/.8) (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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